

Uncooled O-band InP MZ Modulator PIC for 3.2 Tb/s (400 Gb/s/lane) Pluggable Transceiver

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Abstract: We developed a 100-GHz-class bandwidth TEC-less O-band 8-channel InP MZ modulator. The small device exhibits net 3.2-Tb/s (400-Gb/s/lane) 500 m transmission over the range of 20 to 80°C. © 2025 The Author(s)

1. Introduction

Ever-growing data traffic has accelerated the deployment of higher speed electrical and optical devices even in shorter reach communications. In particular, the rapid growth in intra-data centers is becoming an urgent issue to satisfy emerging internet content. Although the next generation 1.6-T and even 3.2-T Ethernet (400 G/lane) are expected to become vital systems within the next five years, as yet there are no established optics by a mature platform. The InP-based electro-absorption modulator integrated laser diode (EA-DFB) and silicon photonics-based (SiP) Mach-Zehnder modulator (MZM) are commonly utilized as the O-band electro-optic modulator for short-reach applications owing to their mature and well-developed platforms [1–4]. However, they still struggle to obtain both the 100-GHz-class bandwidth and a sufficient optical modulation amplitude (OMA) without an external driver IC, which eventually affects the power consumption of the transceiver. Alternatively, thin-film LiNbO₃ (TFLN) MZM shows promise as a platform because of the very smooth roll-off over 100 GHz, low-driving voltage, and manufacturing accessibility [5–9]. Despite their excellent performance, they need a relatively large footprint to keep a low driving voltage. Another candidate is InP-MZM, as it exhibits a high bandwidth and low driving voltage with much smaller footprint, and its mature processes will soon be ready to handle 6-inch huge volume production [10, 11]. Nevertheless, there have been no sufficient reports of O-band InP-MZM yet, possibly because it generally requires a thermoelectric cooler (TEC), which drives up the cost and power requirements.

In this paper, we report for the first time a 100-GHz-class uncooled O-band 8-channel InP-MZM PIC for the next-generation 3.2-Tb/s small form-factor pluggable (SFP) transceiver. By utilizing a proven design of our IQ modulator products, we successfully demonstrated 172-GBd PAM-6 500-m IM/DD transmission with a BER below 3×10^{-3} for all channels, achieving net-3.2 Tb/s (400 Gb/s/lane) over the range of 20 to 80°C.

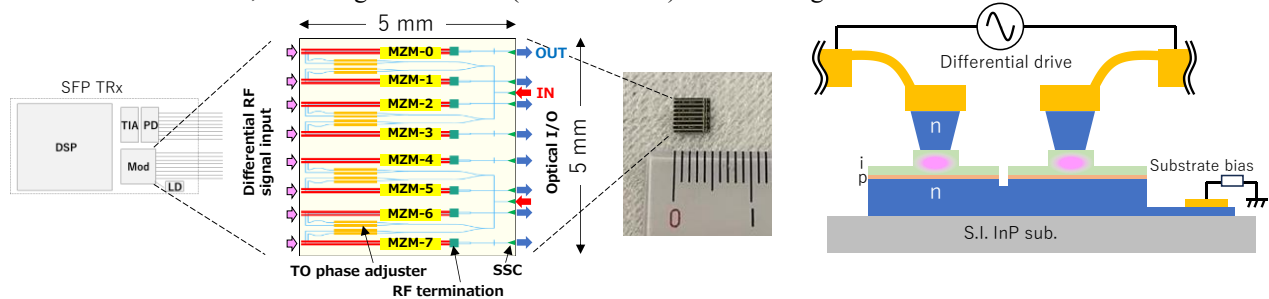


Fig. 1. Schematic illustration, cross-section, and photograph of the modulator PIC.

2. Modulator fundamental performance

Fig. 1 shows a schematic illustration and photograph of the modulator PIC designed for single-wavelength 8-lane parallel transmission such as DR8. Eight MZMs are integrated in a die with a footprint of 5.0×5.0 mm². If we implement a 1×8 configuration with an optical splitting loss of 9 dB, the input power from a single LD would be marginal in terms of optical signal-to-noise ratio (OSNR) requirement for the high symbol rate. Therefore, we utilize the dual 1×4 configuration with two optical input ports. In the modulation region, we design an n-i-p-n heterostructure with low electrical and optical losses [11] in which the photoluminescence (PL) wavelength of the multi-quantum well (MQW) is adjusted considering band-edge absorption at high temperatures and modulation

efficiency over the operating temperature. The designs of the differential RF circuits of MZM, channel pitch (625 μm), on-chip termination resistor (65 Ω differential), thermo-optic (TO) phase adjusters (half-wave shift power: 20 mW), and optical spot-size converter (SSC) are the same as those of our commercial C/L-band twin-IQ modulator [11]. Fig. 2(a) shows the relation between the half-wave voltage (V_π) and additional band-edge optical absorption loss. The V_π is controlled by substrate bias, where 2.0V of V_π is obtained at the cost of an additional loss of 1 dB. The optical insertion loss of each MZM at the maximum transmission point and the V_π of 2.2V is around 12.0 dB, which includes 6-dB intrinsic splitting loss for four MZMs and 4.4-dB lensed-fiber coupling losses (2.2 dB/facet). Fig. 2(b) shows the static extinction characteristics of the 2.0-V V_π condition at temperatures of 20, 50, and 80°C. By adjusting the substrate bias for each temperature, we obtained equivalent transfer curves while keeping an extinction ratio (ER) of over 30 dB. We also confirmed a smooth roll-off of small-signal electro-optic (EO) response, as shown in Fig. 2(c). The EO bandwidth of the chip reaches 100 GHz under all temperature conditions. These characteristics show that the MZM is capable of 224-GBd-class intensity modulation with sub-1V_{ppd} swing voltage.

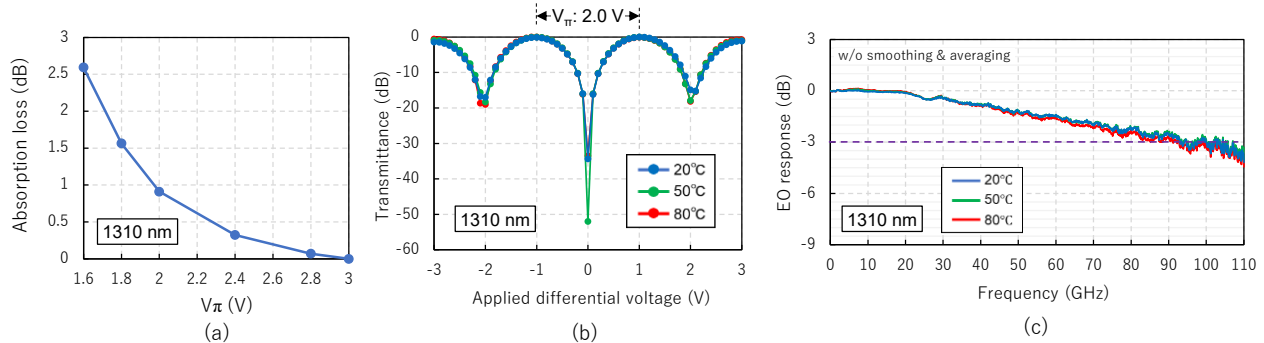


Fig. 2. Static characteristics of MZM: (a) additional absorption loss dependence of V_π , (b) temperature dependence of the MZM transfer curve, and (c) temperature dependence of EO responses.

3. Modulation experiments

Fig. 3(a) shows the experimental setup. The MZM PIC was mounted on a stage capable of setting ambient temperature over the range of 20–80°C and accessed by lensed fibers. We set the substrate bias so that V_π is 2.2V. Each MZM was tested one by one with an RF probe supplying a differential electrical signal from a single channel of a 256-GSa/s 80-GHz-bandwidth arbitrary waveform generator (AWG). We utilized a commercial high-output (+20 dBm) DFB-LD chip as a laser source with a peak wavelength of around 1311 nm. The optical output power from each MZM was around +4 dBm, where 10% of it was tapped off for optical power monitoring to align the lensed fibers and spectrum monitoring, and the rest of it was launched into a 500-m standard single-mode fiber (SMF). At the receiver side, we used a PIN photodiode (PIN-PD) with an opto-electrical (OE) 3-dB bandwidth of 100 GHz. Due to unavailability of a 100-GHz-class transimpedance amplifier (TIA), we inserted a Praseodymium-doped fiber amplifier (PDFA) after the SMF. A variable optical attenuator (VOA) was used to control the received optical power (ROP) launched to the PD. The electrical signal from the PD was captured by a 256-GSa/s 113-GHz-bandwidth digital storage oscilloscope (DSO). As the modulation format, we selected 172-GBaud 2.5-bits/symbol 6-level pulse amplitude modulation (PAM-6), where each symbol pair represents two quadratures of a 32-level quadrature amplitude modulation (32QAM) symbol [12]. The gross bit rate was $172 \times 2.5 = 430$ Gbps. In the transmitter-side offline DSP, the pulse-shaped PAM-6 signal was pre-emphasized to compensate for the frequency responses of the AWG, RF cable, and probe. To mitigate the impact of RF reflection due to the impedance mismatch at the modulator interface (65 Ω differential) and the AWG (100 Ω) connected via the 250-mm-long cable and probe, we also applied a linear T/2-spaced FIR filter with 4001 fixed-weight taps. This filter is expected to be significantly downsized or omitted by assembling the MZM close to a DSP die and/or co-designing their interfaces. In the receiver-side DSP, we used a T/2-spaced 61-tap feed-forward adaptive equalizer (AEQ) before calculating the bit error rate (BER) based on hard decision. We did not use any digital nonlinear compensators in either the transmitter or receiver side.

Fig. 3(b) shows the optical spectrum measured at the MZM output, reflecting its high-bandwidth modulation capability. Fig. 3(c) shows digitally interpolated eye diagrams after the AEQ measured at the ROP of +11 dBm. The six signal levels are clearly distinguished for all the eight channels and the temperatures of 20, 50, and 80°C. In all cases, the BER was below the threshold of the 7%-overhead (OH) hard-decision forward error correction (FEC) code of 3.8×10^{-3} , yielding a net bit rate of 401.8 Gbps/lane. In Fig. 3(d) and (e), BER vs. ROP for different

temperatures and MZM channels, respectively, are shown. The BER performance was mostly insensitive to the temperature and was uniform over the eight channels, having the ROP sensitivity of around +8 dBm with the TIA-less PD. In addition, we confirmed that the far-end crosstalk characteristic between adjacent channels was less than -30 dB up to 90 GHz, indicating that there would be no significant BER degradation even if all eight channels were driven simultaneously. From these results, the PIC is expected to be capable of TEC-less 3.2-Tbps operation over a wide range of ambient temperatures.

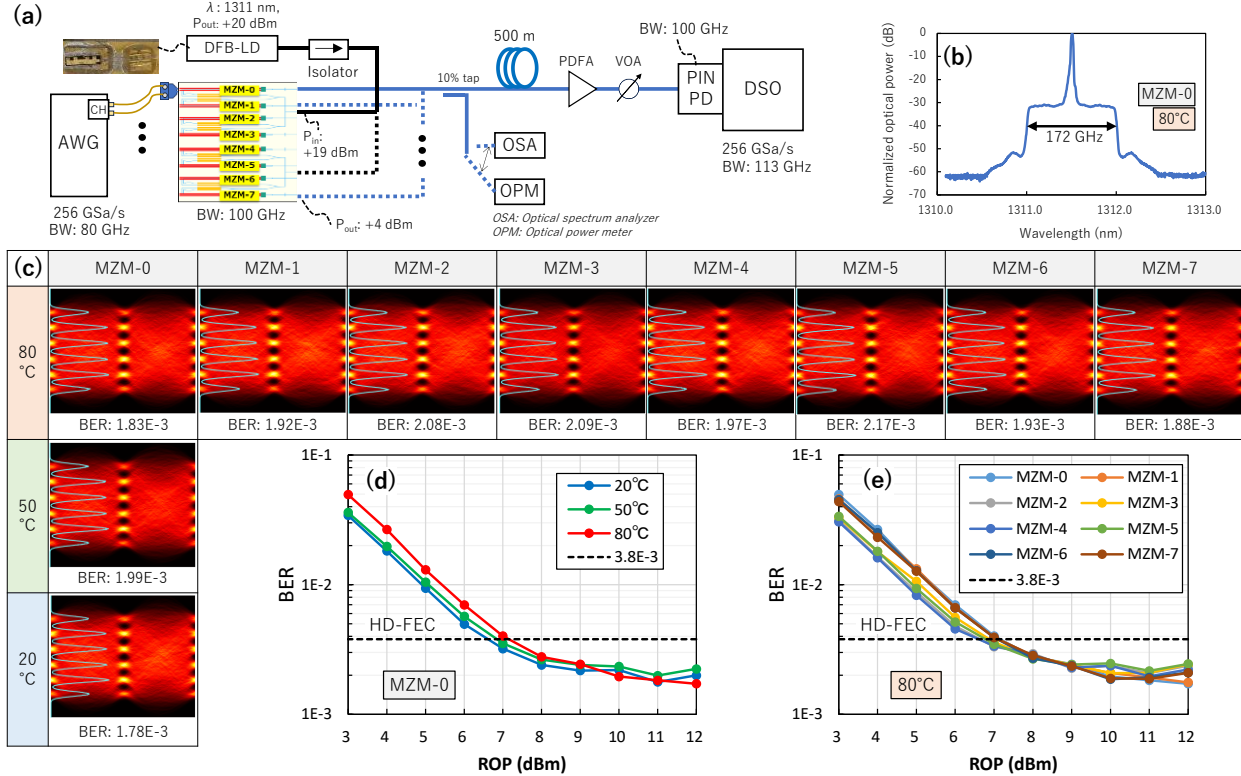


Fig. 3. (a) Experimental setup for 172-GBd PAM-6 500-m transmission. (b) Optical spectrum after modulator output. (c) Temperature and channel dependence of recovered eye at ROP of 11 dBm, and BER v.s. ROP. (d) Temperature dependence. (e) Channel dependence.

4. Conclusion

We developed a 100-GHz-class uncooled 8-channel O-band InP-MZM PIC that exhibited net-3.2-Tb/s 500-m transmission over the temperature range from 20 to 80°C. The small-footprint and low-driving-voltage MZM PIC will be key for the next-generation 1.6-T DR4 and 3.2-T DR8 SFP transceivers with PAM-DSP direct driving.

Acknowledgement

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